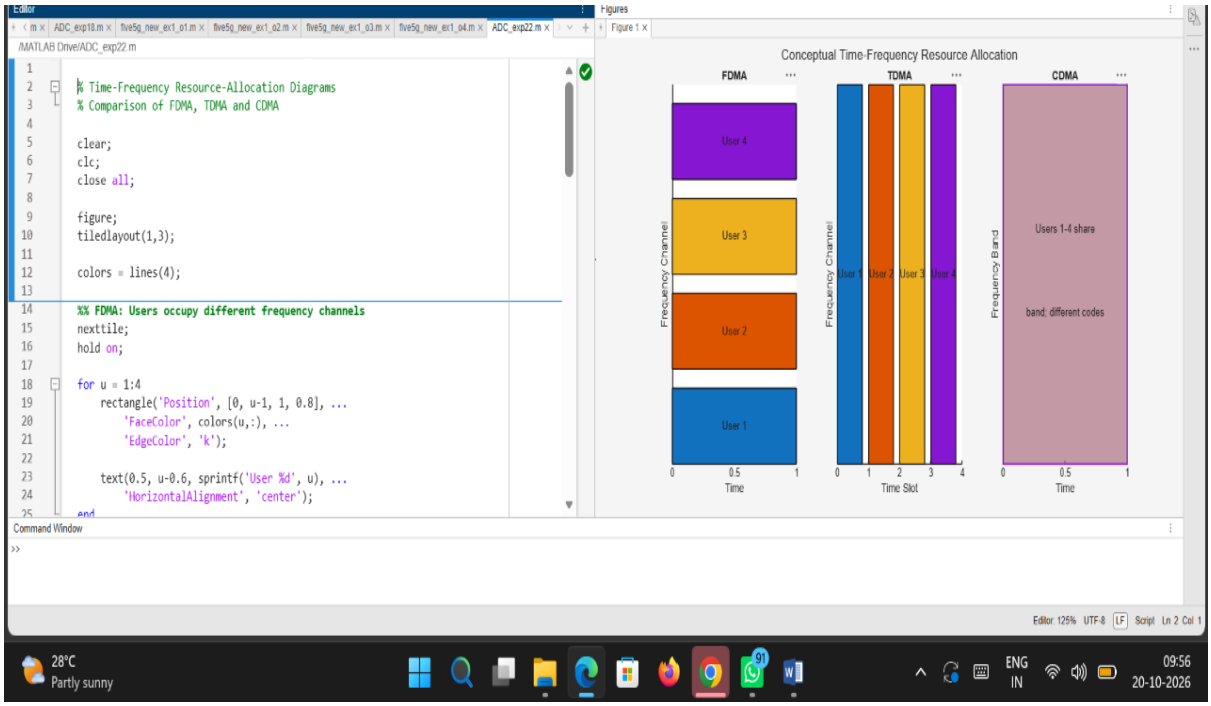
Ex.No:22: SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

Objective 1:



Code:

```
% Time-Frequency Resource-Allocation Diagrams
% Comparison of FDMA, TDMA and CDMA

clear;
clc;
close all;

figure;
tiledlayout(1,3);

colors = lines(4);

%% FDMA: Users occupy different frequency channels
nexttile;
hold on;

for u = 1:4
    rectangle('Position', [0, u-1, 1, 0.8], ...
        'FaceColor', colors(u,:), ...
        'EdgeColor', 'k');

    text(0.5, u-0.6, sprintf('User %d', u), ...
        'HorizontalAlignment', 'center');
```

```
end

xlim([0 1]);
ylim([0 4]);
xlabel('Time');
ylabel('Frequency Channel');
title('FDMA');
set(gca, 'YTick', []);
hold off;

%% TDMA: Users occupy different time slots
nexttile;
hold on;

for u = 1:4
    rectangle('Position', [u-1, 0, 0.8, 1], ...
        'FaceColor', colors(u,:), ...
        'EdgeColor', 'k');

    text(u-0.6, 0.5, sprintf('User %d', u), ...
        'HorizontalAlignment', 'center');
end

xlim([0 4]);
ylim([0 1]);
xlabel('Time Slot');
ylabel('Frequency Channel');
title('TDMA');
set(gca, 'YTick', []);
hold off;

%% CDMA: Users share the same time and frequency band
nexttile;
hold on;

for u = 1:4
    rectangle('Position', [0, 0, 1, 1], ...
        'FaceColor', colors(u,:), ...
        'FaceAlpha', 0.22, ...
        'EdgeColor', colors(u,:), ...
        'LineWidth', 1.5);
end

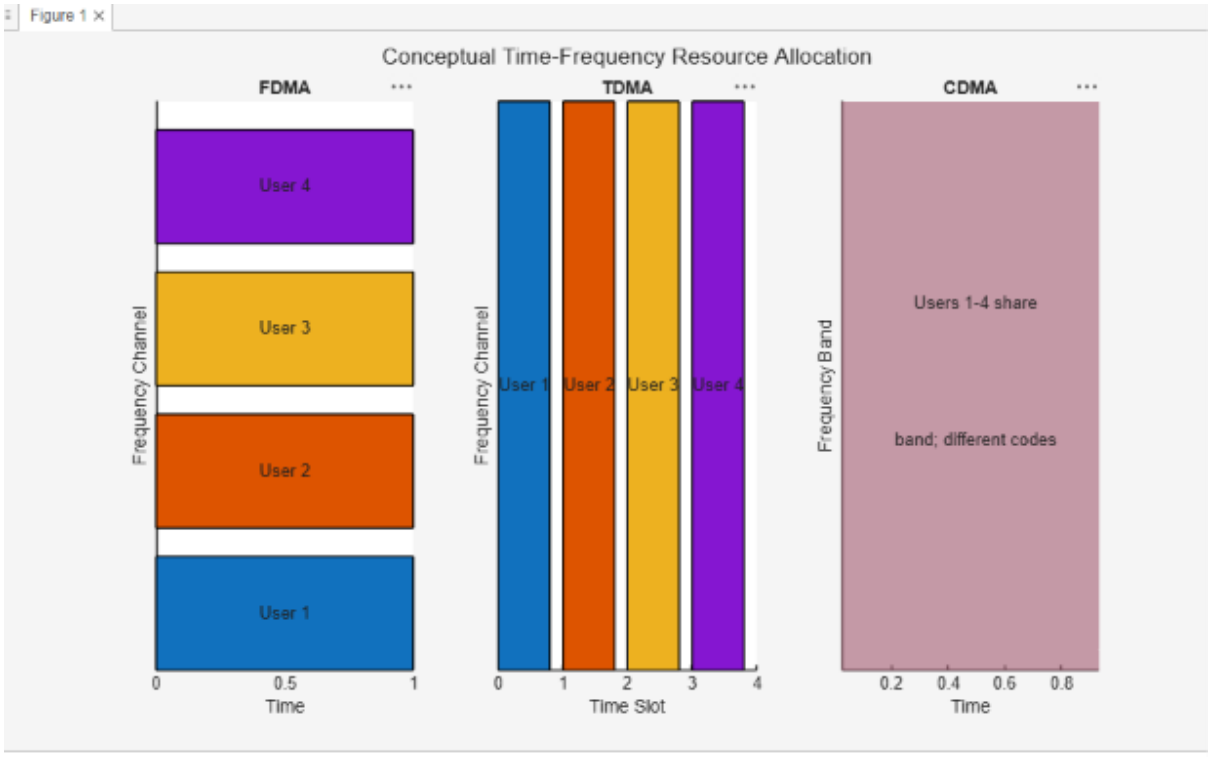
text(0.5, 0.62, 'Users 1-4 share', ...
    'HorizontalAlignment', 'center');

text(0.5, 0.40, 'band; different codes', ...
    'HorizontalAlignment', 'center');

xlim([0 1]);
ylim([0 1]);
xlabel('Time');
ylabel('Frequency Band');
title('CDMA');
set(gca, 'YTick', []);
hold off;

%% Overall Figure Title
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

Output:


Figure 1
Conceptual Time-Frequency Resource Allocation
FDMA
User 4
User 3
User 2
User 1
Frequency Channel
0
0.5
1
Time
TDMA
User 1
User 2
User 3
User 4
Frequency Channel
0
1
2
3
4
Time Slot
CDMA
Users 1-4 share
band; different codes
Frequency Band
0.2
0.4
0.6
0.8
Time